E14

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1 Lab Report: Semiconductor Diodes (E14e)

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1.2 1. Theoretical Background

Semiconductor diodes are key components in modern electronics. They are p-n junctions that exhibit nonlinear current-voltage (I–V) characteristics depending on the biasing condition.

1.2.1 1.1 Forward Bias Region

In forward bias, a diode allows current flow once the forward voltage exceeds the threshold voltage (typically $0.6–0.7V$ for silicon, $2–3V$ for LEDs). The current rises exponentially with voltage:

$$I = I_0(e^{\frac{qU}{nkT}} - 1)$$

Where: - I is the diode current. - U is the voltage across the diode. - q is the electron charge. - k is the Boltzmann constant. - n is the ideality factor. - T is the absolute temperature.

1.2.2 1.2 Reverse Bias Region

In reverse bias, a small saturation current flows until breakdown occurs (in Zener diodes), characterized by a sharp increase in reverse current at the breakdown voltage.

1.2.3 1.3 LED Emission

For LEDs, the voltage drop correlates with the photon energy $E = h\nu = qUE = h\nu = qU$, allowing estimation of the bandgap energy.

1.2.4 1.4 Junction Capacitance

In a reverse-biased p-n junction, the depletion region acts as a capacitor. The depletion capacitance C decreases with reverse voltage U, following:

$$\frac{1}{C^2} \propto U$$

This relation allows determination of the doping concentration from the slope of a $1/C^2$ vs U plot.

1.3 2. Experimental Setup

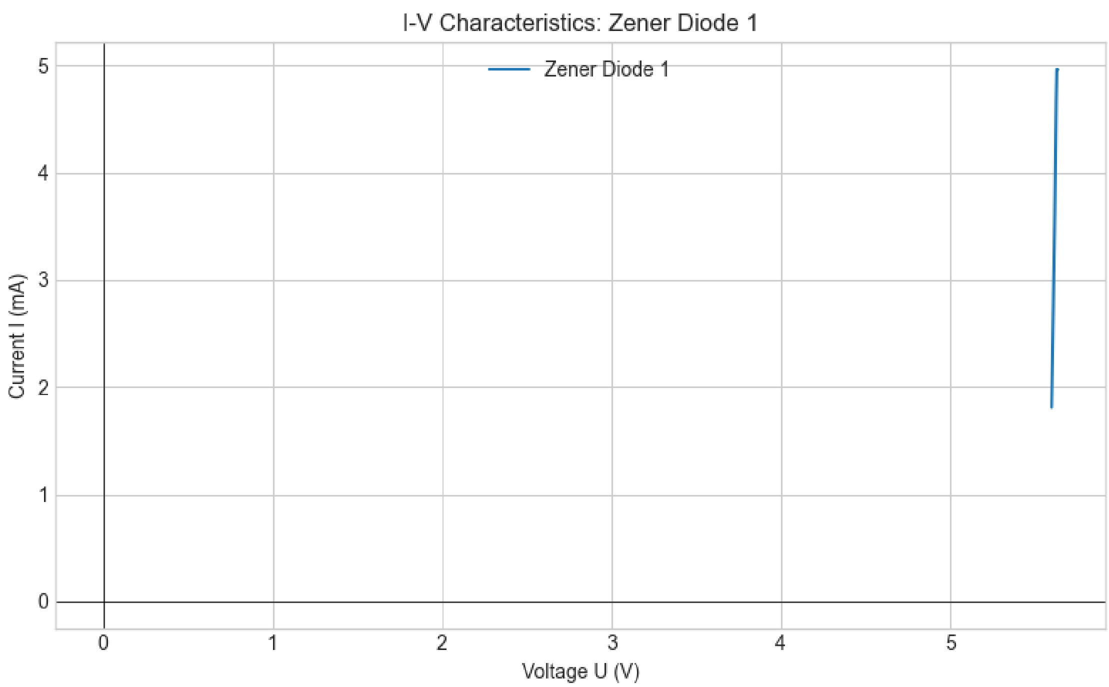
- Diodes tested: Silicon diode (Si), Light Emitting Diode (LED), Zener diode (Zener1 and Zener2).
- Measurements taken:
  - Current I vs Voltage U characteristics for forward and reverse bias
  - Capacitance vs reverse voltage using an LCR meter
- Equipment:
  - Source meter for I-V measurements
  - LCR meter for capacitance
  - Oscilloscope

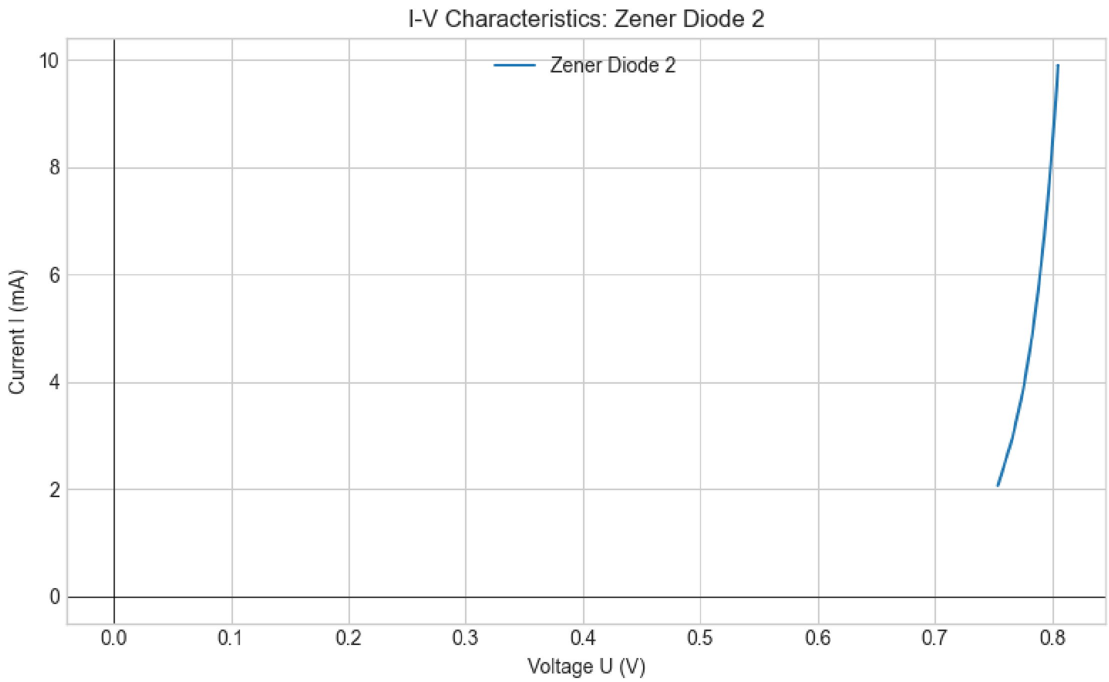
Data was logged digitally in CSV and Excel formats.

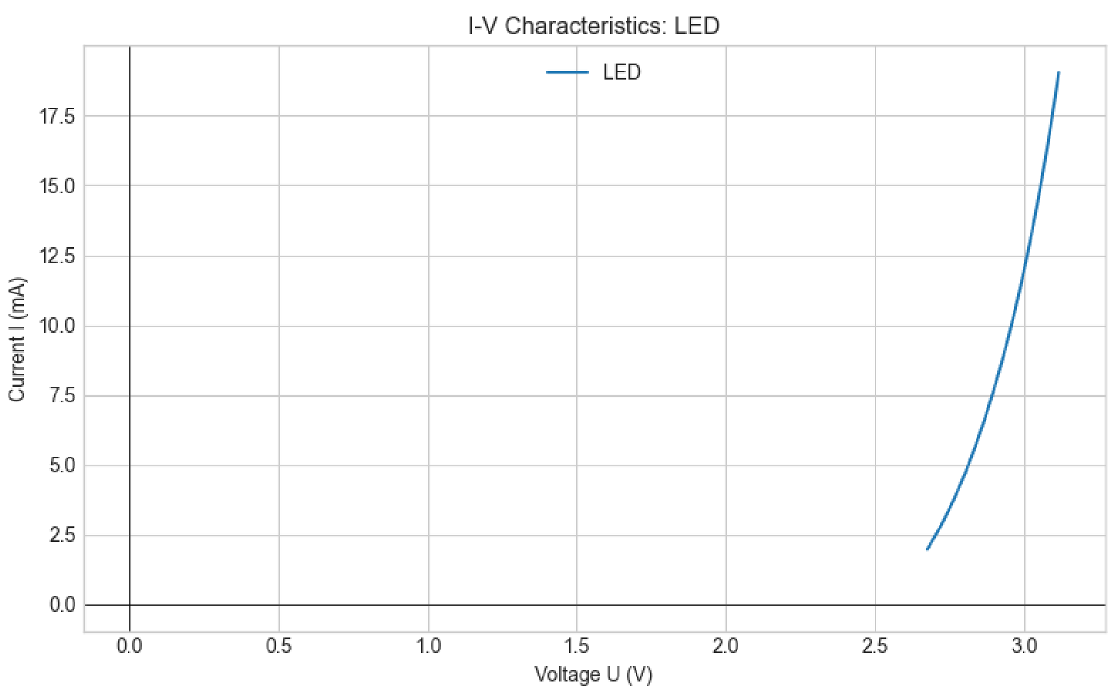
1.4 3. Data Analysis & Python Code

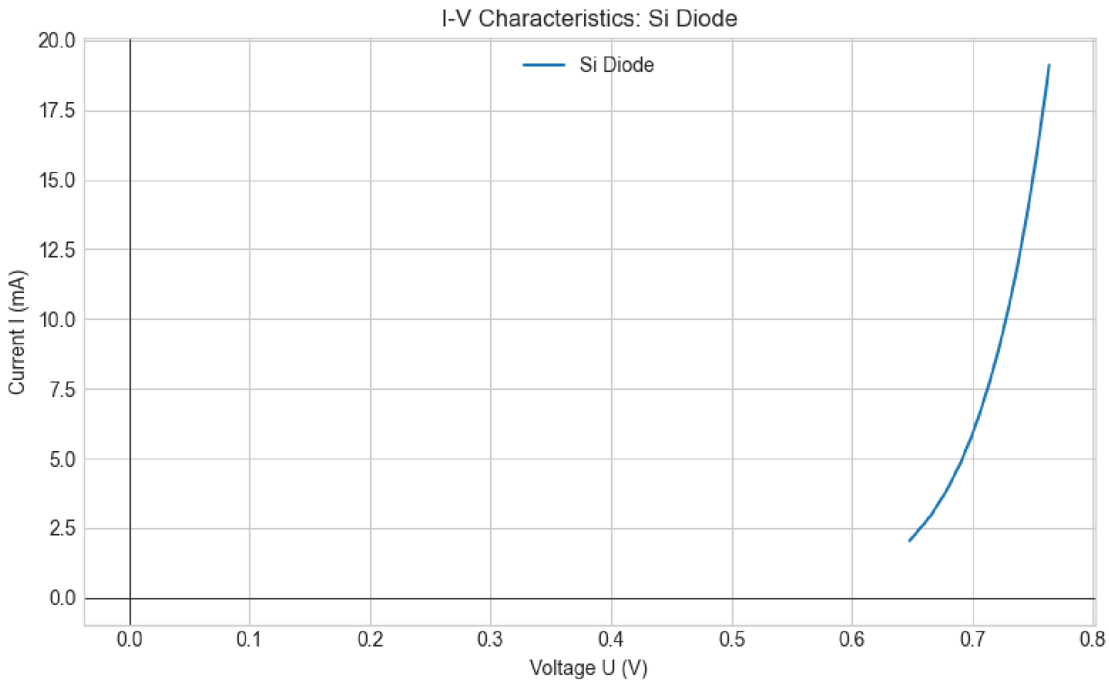
1.4.1 3.1 Task 1: I–V Characteristics (Zener1, Zener2, LED, Si)

Python code for loading and plotting:




I-V Characteristics: Zener Diode 2
Zener Diode 2
Current I (mA)
Voltage U (V)
0
2
4
6
8
10
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


I-V Characteristics: LED
LED
Current I (mA)
Voltage U (V)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0



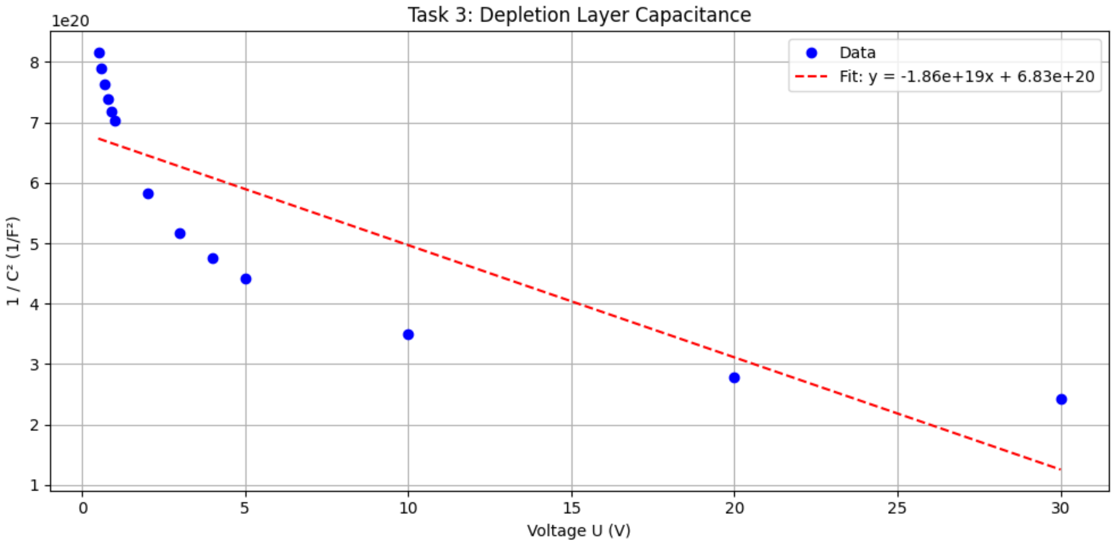
This produces a clear plot showing: - Exponential current increase for Si and LED in forward bias - Threshold voltages (~0.7V for Si, ~2.6–3.1V for LED) - Zener breakdown in reverse bias (around 5.6V for Zener1, ~0.8V for Zener2)

1.4.2 3.2 Task 2: LED Bandgap Estimation

We estimate the bandgap energy from the LED forward voltage $U_{on} \approx 2.6–3.1V$:

$$E_g \approx qU_{on} \rightarrow E_g \approx (1.602 \times 10^{-19}C) \times 3.0V = 3.0eV$$

1.4.3 3.3 Task 3: Depletion Layer Capacitance

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Linear regression results:
Slope     = -1.857e+19  1/F²·V
Intercept = 6.825e+20  1/F²
R²        = 0.7008
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